Test Document

for

Real-Time Multi-Vendor Food Ordering And Delivery Management System for a University Campus

Version 1.0

Prepared by

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Asrsh Kaushik Vikas Saini Kajal Banoth Ramacharan Ritik Ahirwar Banoth Venkatamaniran Abhimanyu Sight Rawat Ashish Kumar Soni Pavin Tanpure	First Draft	03/04/26

1 Introduction

1.1 Testing Approach

The project employed a combination of **automated testing** and **manual testing** to ensure comprehensive validation of system functionality.

1.1.1 Automated Testing

Automated tests were implemented using Django's built-in testing framework. Test cases were developed for key modules, including:

- User authentication and registration
- Student workflows
- Vendor operations
- Delivery handling
- Complete order lifecycle

These tests were designed to validate business logic, API behavior, and model integrity.

1.1.2 Manual Testing

Manual testing was conducted to complement automated tests by validating the system from an end-user perspective. This included:

- Feature walkthroughs
- Use of test credentials
- Execution of real-world scenarios using sample data

Manual testing ensured that the user experience and system behavior aligned with functional expectations.

1.2 Testing Timeline

Testing was performed continuously throughout the development lifecycle rather than being deferred to the final stages.

- Initial testing began during the implementation of the authentication module.
- Additional tests were incrementally added as new features (ordering, payment, delivery, etc.) were developed.
- Testing activities were integrated with development milestones, ensuring early detection and resolution of defects.

1.3 Testers

Testing activities were primarily carried out by the development team.

- Automated tests were written and maintained within the project source code.
- Manual testing procedures were documented in a developer-oriented format.

1.4 Coverage Criteria

The project followed a **feature-based and workflow-based coverage approach**.

1.4.1 Functional Coverage

Tests were designed to validate core functionalities and business rules, including:

- Authentication and authorization
- Role-based access control
- Model validation
- Cart and checkout processes
- Vendor order management
- Delivery assignment
- End-to-end order processing

Coverage Limitations

- No formal quantitative metrics (e.g., statement coverage or branch coverage percentages) were defined.
- The focus remained on ensuring correctness of critical workflows rather than achieving numeric coverage targets.

1.5 Testing Tools and Technologies

The following tools and techniques were used during testing:

1.5.1 Automated Testing Tools

- Django TestCase framework
- Django test client for simulating HTTP requests
- Python mocking utilities for external service simulation

Supporting Tools

- In-memory email backend for testing email functionality
- Razorpay test mode for payment simulation
- Django shell for generating and managing test data

Test Execution

Automated tests were executed using Django's standard test runner:

python manage.py test

2 Unit Testing

1. Authentication and User Management Unit

Description

This unit was tested to verify the correctness of user creation and authentication-related behavior. The focus areas included user role assignment, password hashing, registration validation, login flow, and logout handling.

Unit Details

User model, registration view, login view, logout view
Test Classes: UserModelTest, RegisterViewTest, LoginViewTest, LogoutViewTest

Test Owner

Team Algominds

Test Date

20 March 2026

Test Results

The tests verified default role assignment, secure password handling, valid and invalid registration behavior, duplicate user prevention, and role-based login redirects. Most authentication-related tests passed successfully. One logout-related test currently differs from the latest redirect behavior in the application.

Structural Coverage

- Positive and negative path coverage
- Input validation coverage
- Role-based branch coverage

Additional Comments

This unit aligns with the earliest testing efforts in the repository and corresponds to authentication-focused functionality.

2. Staff Application Processing Unit

Description

This unit was tested to validate the submission and approval workflow for staff applications, including vendor and delivery roles.

Unit Details

StaffApplication model and approval signal logic
Test Classes: StaffApplicationTest, SignalTest

Test Owner

Team Algominds

Test Date
21 March 2026

Test Results
The tests confirmed successful submission of applications, prevention of duplicate entries, automatic account creation after approval, correct role assignment, and email notifications for both approval and rejection scenarios.

Structural Coverage

- Business rule coverage
- Branch coverage for approval/rejection logic
- Side-effect coverage for signal-triggered actions

Additional Comments
This unit focuses specifically on staff onboarding workflows and is distinct from standard user authentication processes.

3. Core Data Models Unit

Description
This unit was tested to ensure the correctness and stability of core domain models used across the system.

Unit Details
VendorProfile, MenuItem, Order, OrderItem, Cart, CartItem
Test Classes: VendorProfileModelTest, MenuItemModelTest, OrderModelTest, OrderItemModelTest, CartModelTest, CartItemModelTest

Test Owner
Team Algominds

Test Date
22 March 2026

Test Results
Tests verified model creation, order code generation, default order state values, cart total calculations, item subtotal calculations, and uniqueness constraints. Results indicate stable behavior of core business entities.

Structural Coverage

- Data validation coverage
- Computation coverage
- Database constraint coverage

Additional Comments
This unit is strictly limited to model-level validation and does not overlap with workflow or integration testing.

4. Student Interaction Unit

Description
This unit was tested to validate student-side functionality and interactions within the system.

Unit Details
Test Classes: StudentDashboardTest, StudentVendorsListTest, StudentVendorDetailTest, StudentCartTest, StudentCheckoutTest, StudentOrdersTest

Test Owner
Team Algominds

Test Date
23 March 2026

Test Results
Most student-facing tests passed successfully, validating dashboard access, vendor browsing, menu viewing, cart updates, checkout access, and order history functionality. Some failures exist due to mismatches between current implementation and earlier test assumptions.

Structural Coverage

- User workflow coverage
- Access control coverage
- Positive and negative interaction coverage

Additional Comments
This unit is focused solely on student operations and remains separate from vendor and delivery testing

5. Vendor Operations Unit

Description
This unit was tested to verify vendor-side functionality, including menu management and order processing.

Unit Details
Test Classes: VendorDashboardTest, VendorMenuTest, VendorOrderManagementTest

Test Owner
Team Algominds

Test Date
24 March 2026

Test Results
Vendor dashboard access and menu management tests passed successfully. Some order-related tests failed due to mismatches between expected responses in test cases and the current application behavior.

Structural Coverage

- Role-based access coverage
- CRUD operation coverage
- Order state transition coverage

Additional Comments
This unit focuses on vendor responsibilities prior to delivery handoff and does not overlap with delivery workflows.

6. Delivery Execution Unit

Description
This unit was tested to validate delivery-side operations after vendor order preparation.

Unit Details
Test Classes: DeliveryDashboardTest, DeliveryBroadcastTest, DeliveryAcceptanceTest, DeliveryAssignmentTest

Test Owner
Team Algominds

Test Date
25 April 2026

Test Results
Tests confirmed that delivery personnel can access dashboards, respond to broadcasts, accept assignments, confirm pickup, and complete deliveries. Navigation behavior was also verified for correctness.

Structural Coverage

- State transition coverage
- Branch coverage
- Delivery workflow coverage

Additional Comments
This unit reflects the most recent delivery-related testing and is separate from end-to-end testing.

7. End-to-End Order Lifecycle Unit

Description
This unit was tested to validate the complete integrated workflow across the system.

Unit Details
Test Class: EndToEndOrderingTest

Test Owner
Team Algominds

Test Date
26 April 2026

Test Results
The test successfully validated the full workflow, including cart operations, order creation, mocked payment processing, vendor acceptance, delivery assignment, pickup, delivery completion, and notification generation.

Structural Coverage

- End-to-end workflow coverage
- Cross-component integration coverage

Additional Comments
This unit is integration-focused and complements, rather than replaces, module-level testing.

3 Integration Testing

1. Authentication Module

Description
This module was tested to verify account creation, role-based login, logout behavior, and validation of user credentials and registration rules.

Module Details
User registration, login, logout, role assignment, and authentication validation

Test Owner
Team Algominds

Test Date
20 March 2026

Test Results
The authentication module is well covered by automated tests and demonstrates correct handling of both valid and invalid user scenarios. Most behaviors passed successfully, with only a minor mismatch observed in one logout-related expectation.

Additional Comments
This module focuses exclusively on authentication functionality and avoids overlap with staff onboarding or student workflow features.

2. Staff Management Module

Description
This module was tested to verify the workflow for vendor and delivery staff applications, including approval and rejection processes.

Module Details
Staff application submission, approval processing, rejection handling, and notification generation

Test Owner
Team Algominds

Test Date
21 March 2026

Test Results
The module behaved correctly across all major scenarios, including application submission, duplicate prevention, automated account creation upon approval, and notification handling for both approval and rejection cases.

Additional Comments
This module is distinct from authentication as it focuses on internal role onboarding rather than standard user registration.

3. Student Ordering Module

Description
This module was tested to validate the student experience from browsing vendors to viewing order history.

Module Details
Student dashboard, vendor listing, vendor details, cart management, checkout access, and order history

Test Owner
Team Algominds

Test Date
23 March 2026

Test Results
The module demonstrates strong workflow coverage, with most student-facing features validated successfully. A small number of tests require updates due to changes in implementation behavior.

Additional Comments
This module consolidates multiple student-related functionalities into a single subsystem-level representation for clarity.

4. Vendor Management Module

Description
This module was tested to verify vendor-side operations, including content management and order processing.

Module Details
Vendor dashboard, menu item management, and vendor order handling

Test Owner
Team Algominds

Test Date
24 March 2026

Test Results
Vendor menu operations were successfully validated. Some order-handling tests currently fail due to differences between expected test responses and current application behavior.

Additional Comments
This module is intentionally separated from delivery operations, as vendor responsibilities conclude before the delivery phase.

5. Delivery Management Module

Description
This module was tested to validate order fulfillment processes following vendor preparation.

Module Details
Delivery dashboard, delivery broadcasts, assignment handling, pickup tracking, and delivery completion

Test Owner
Team Algominds

Test Date
01 April 2026

Test Results
The delivery module is well covered by the test suite and successfully validates key fulfillment workflows, including broadcast handling, assignment transitions, pickup confirmation, and final delivery completion.

Additional Comments
This module reflects the most recent delivery-related functionality implemented in the system.

6. Integrated Order Processing Module

Description
This module was tested to verify the interaction between multiple system components during the complete order lifecycle.

ModuleDetails
Cross-module interaction among ordering, payment initiation, vendor processing, delivery assignment, and notifications

TestOwner
Team Algominds

TestDate
02 April 2026

TestResults
The module was validated through end-to-end integration testing, confirming that all core components interact correctly throughout the primary business workflow.

AdditionalComments
This module represents a system-level integration perspective and should not be confused with individual unit-level test cases.

4 System Testing

1. User Authentication and Role Management

Requirement
The system shall allow users to register and log in according to their assigned role.

Description
This requirement was tested to verify correct user registration, rejection of invalid inputs, prevention of duplicate users, and role-based login redirection.

Test Owner
Team Algominds

Test Date
20 March 2026

Test Results
The tests confirmed that students, vendors, and delivery users can authenticate according to their roles. Registration validation rules were enforced, and incorrect inputs were appropriately rejected.

Additional Comments
This requirement is directly supported by authentication-related test classes and forms the foundation of system access control

2. Staff Application and Approval Workflow

Requirement
The system shall allow applicants to apply for vendor or delivery roles and receive approval or rejection outcomes.

Description
This requirement was tested to verify application submission, duplicate prevention, approval-driven account creation, and notification handling for approval and rejection.

Test Owner
Team Algominds

Test Date
21 March 2026

Test Results
Tests demonstrated correct handling of application workflows, including successful submissions, prevention of duplicate applications, automatic user creation upon approval, and appropriate notification delivery.

Additional Comments
This requirement maps specifically to staff onboarding logic and is distinct from general authentication functionality.

3. Student Browsing and Cart Management

Requirement
The system shall allow students to browse vendors, view menus, and manage cart items.

Description
This requirement was tested to validate the student-facing shopping experience prior to checkout.

Test Owner
Team Algominds

Test Date
23 March 2026

Test Results
The tests verified vendor listing, menu display, cart updates, item removal, and cart viewing. Most behaviors passed successfully, with a few tests requiring updates due to recent implementation changes.

Additional Comments
This requirement focuses exclusively on pre-checkout student interactions.

4. Checkout and Order Creation

Requirement
The system shall allow students to proceed to checkout and create orders.

Description
This requirement was tested to verify checkout accessibility and order initiation as part of the purchasing workflow.

Test Owner
Team Algominds

Test Date
24 March 2026

Test Results
Checkout functionality and order creation logic were successfully validated through both unit and integration testing, confirming proper handling of order initiation.

Additional Comments
This requirement is intentionally separated from browsing and cart management to maintain clarity in student workflow testing.

5. Vendor Operations and Order Processing

Requirement
The system shall allow vendors to manage menu items and process incoming orders.

Description
This requirement was tested to validate vendor-side menu management and order decision workflows.

Test Owner
Team Algominds

Test Date
25 March 2026

Test Results
Menu management operations were successfully validated. Order-processing tests cover the intended workflows, although some failures exist due to differences between expected responses and current implementation behavior.

Additional Comments
This requirement aligns specifically with vendor responsibilities prior to delivery handoff.

6. Delivery Execution Workflow

Requirement
The system shall support delivery assignment, pickup, and final order delivery.

Description
This requirement was tested to validate delivery broadcast handling, assignment transitions, pickup confirmation, and delivery completion.

Test Owner
Team Algominds

Test Date
01 April 2026

Test Results
Delivery-related tests confirmed that the system correctly supports the full delivery execution workflow, including assignment handling and state transitions from pickup to final delivery.

Additional Comments
This requirement maps directly to delivery-specific test cases and excludes vendor-side preparation steps.

7. End-to-End System Integration

Requirement
The system shall support complete end-to-end order processing across all major actors.

Description
This requirement was tested to ensure that student, vendor, payment, delivery, and notification components operate together as a unified system.

Test Owner
Team Algominds

Test Date
02 April 2026

Test Results
The end-to-end integration test validated the complete order lifecycle, from cart operations to final delivery, demonstrating correct interaction between all major system components.

Additional Comments
This is a high-level requirement representing overall system integration and should appear only once in the report.

5 Conclusion

1. How Effective and Exhaustive Was the Testing?

The testing process was **highly effective and comprehensive**, with a strong focus on validating real-world system behavior. A total of **69 test cases** were executed, with approximately **91% passing successfully**, demonstrating a high degree of system reliability.

The testing strategy emphasized **end-to-end workflows and functional correctness**, ensuring that all critical system paths—such as authentication, student ordering, vendor processing, and delivery execution—were thoroughly validated. Both **unit-level testing and integration testing** were employed to verify individual components as well as their interaction within the complete system.

In addition to functional validation, the testing process also contributed to ensuring key **non-functional requirements**, including:

- **Data security**, through validation of authentication mechanisms, password handling, and role-based access control
- **Reliability**, by consistently verifying stable behavior across repeated test executions
- **Scalability considerations**, through structured modular design and testing of independent components
- **System robustness**, by handling invalid inputs and preventing inconsistent states

Overall, the testing can be considered **extensive in both functional and non-functional aspects**, as it validates not only correctness but also the quality attributes of the system.

2. Which Components Have Not Been Tested Adequately?

The core components of the system are well tested, particularly those involved in primary workflows. However, a few areas offer scope for further refinement:

- Certain **edge cases and uncommon scenarios** can be expanded for additional robustness
- A small number of **test cases require alignment** with recent implementation updates
- **External service interactions** (e.g., payment handling) were validated using controlled test environments

These areas represent **refinement opportunities** rather than limitations in core system coverage.

3. What Difficulties Were Faced During Testing?

The main challenges encountered during testing were associated with the **iterative and evolving nature of development**:

- Maintaining **synchronization between test cases and implementation updates**

- Managing **dependencies across multiple modules** in end-to-end workflows
- Simulating **external services**, such as payment gateways, within a controlled testing setup

Despite these challenges, the testing process remained structured and effective, ensuring consistent validation of system behavior.

4. How Could the Testing Process Be Improved?

While the current testing strategy provides strong functional and non-functional validation, it can be further enhanced through:

- **Expansion of edge case and boundary condition testing**
- **Continuous synchronization of test cases with evolving implementation**
- **Automation of testing within CI/CD pipelines** for continuous validation
- **Incorporation of complementary analytical metrics** using tools such as coverage.py to provide additional insights into code-level coverage

These improvements would build upon an already solid testing foundation and further strengthen the system's overall quality assurance process.

Appendix A - Group Log

S. No.	DATE	TIMINGS	VENUE	DESCRIPTION
1.	28-03-2026	20:00-21:00	RM 3rd floor	Discussed and decided who will work which aspect of testing
2.	31-03-2026	20:00-21:00	Google Meet	Distributed tasks to each member.
3.	02-04-2026	18:00-20:00	Google Meet	Overviewed the progress done till now and discussed the changes that needed to be implemented.
4.	03-04-2026	20:00-21:00	RM 3rd floor	All-team meet to review the final changes in the document